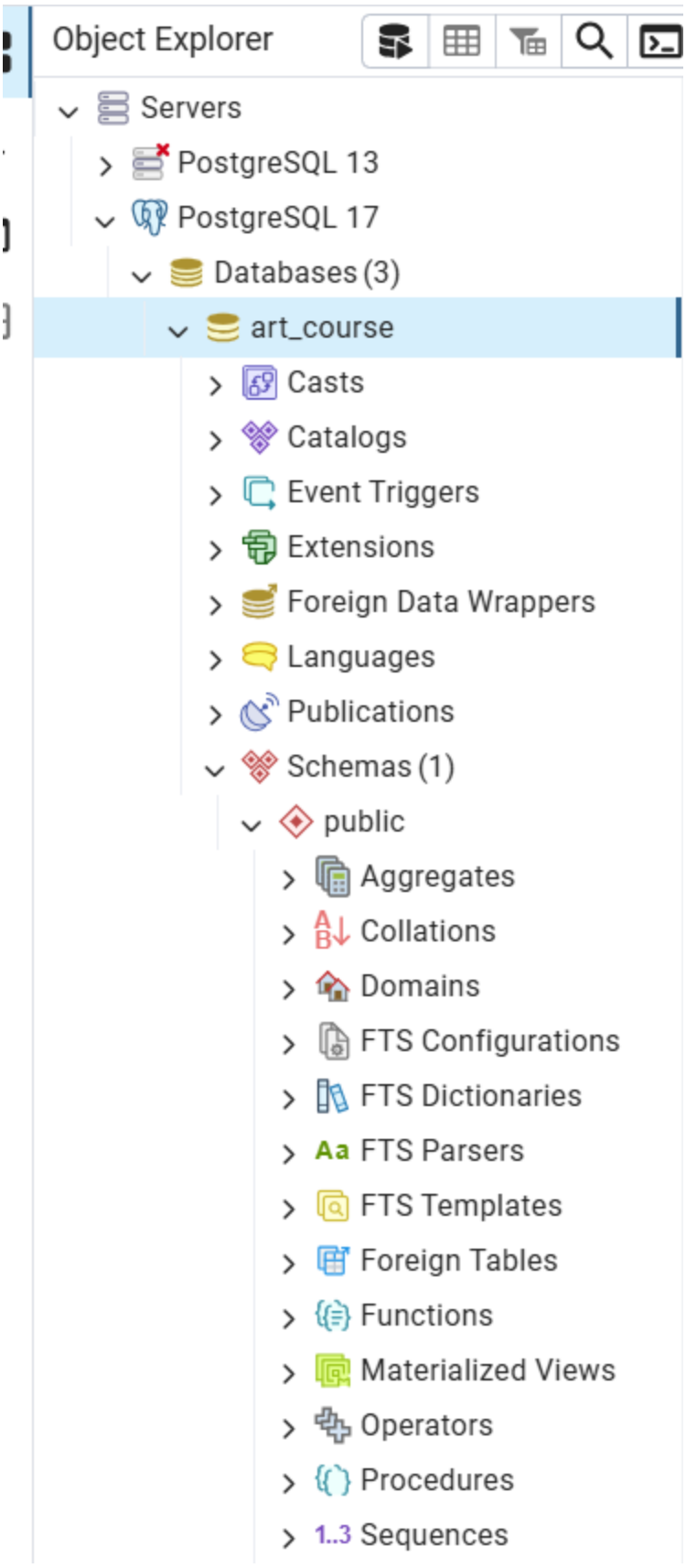
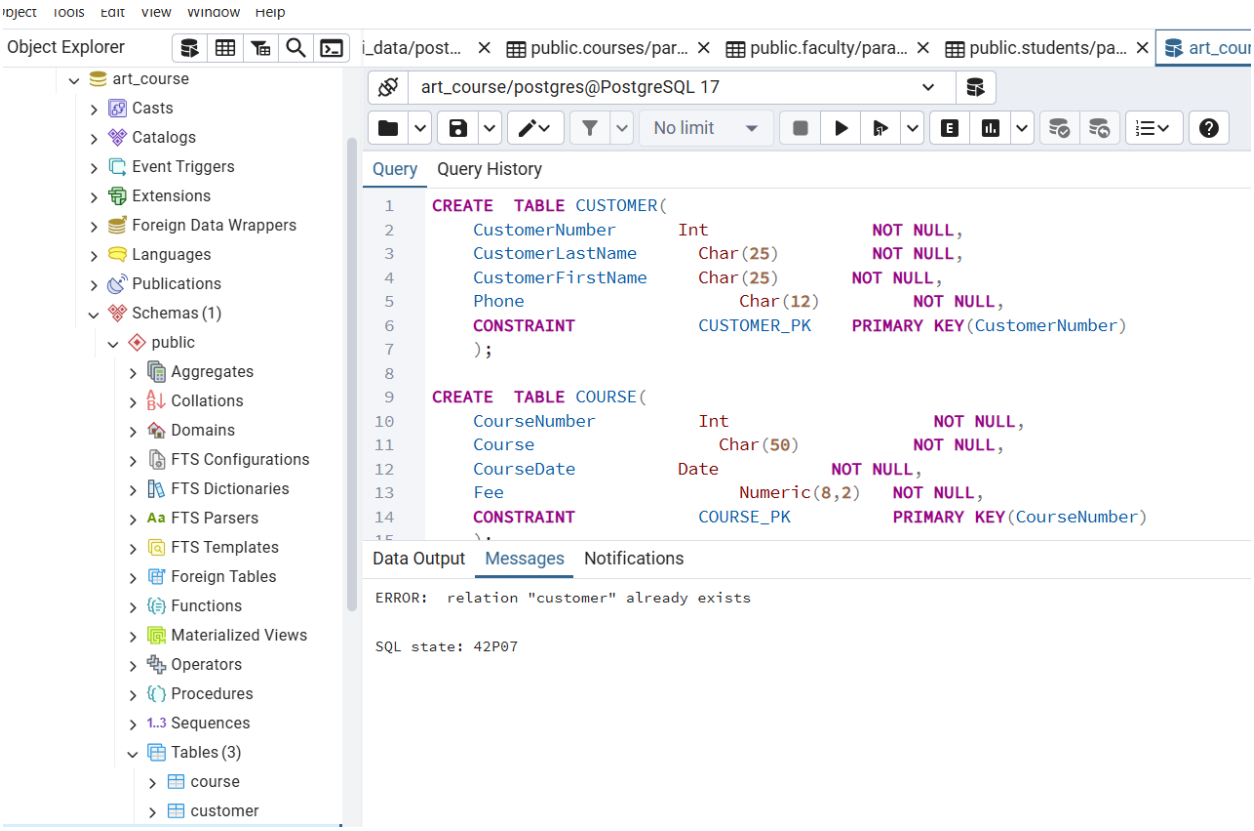
Name: Min Thant Hein

Student-ID: PIUS20230001

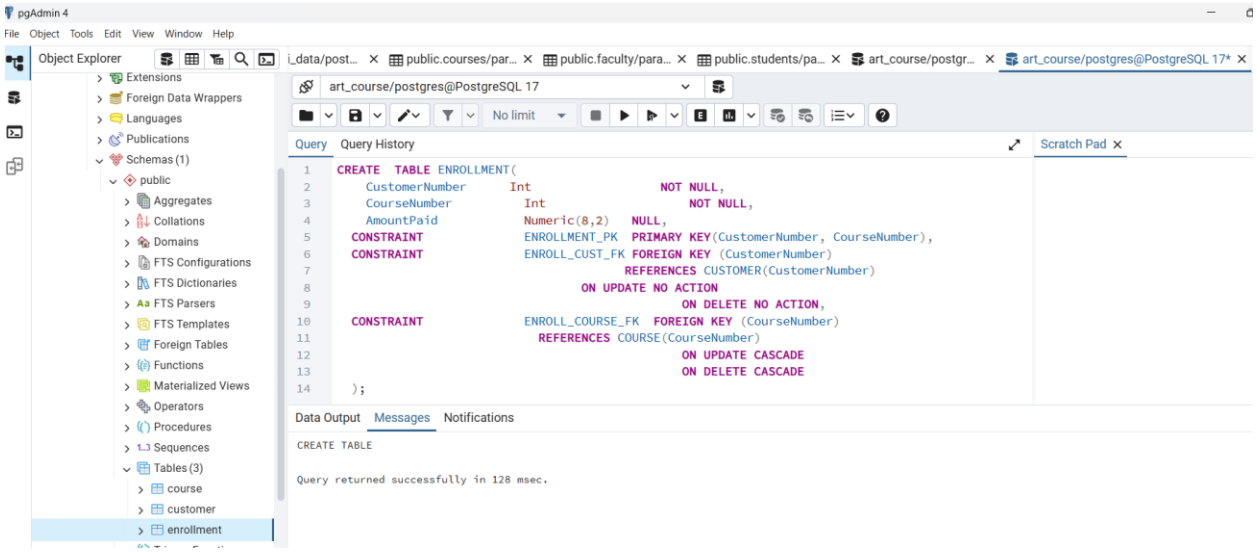
1.Create Art_Course


Object Explorer
Servers
PostgreSQL 13
PostgreSQL 17
Databases (3)
art_course
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences

2.Create Customer Table and Course Table



3. Create Enrollment Table:



4. Inserting the data and table views

Customer table:

Query:

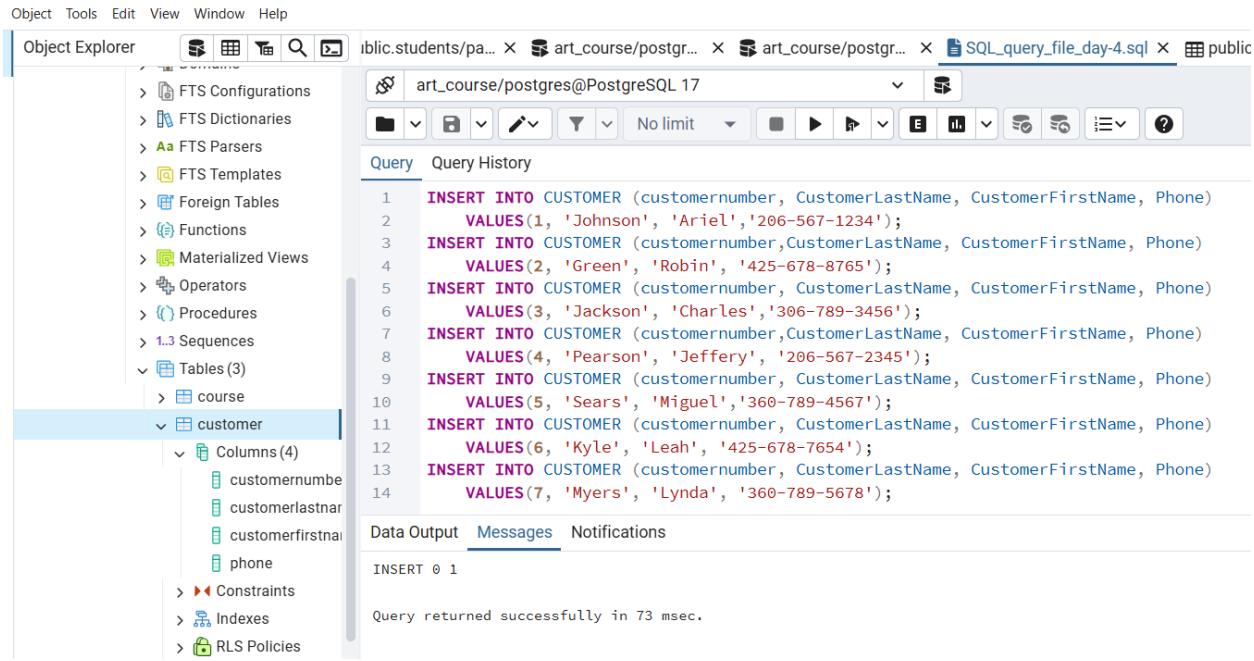
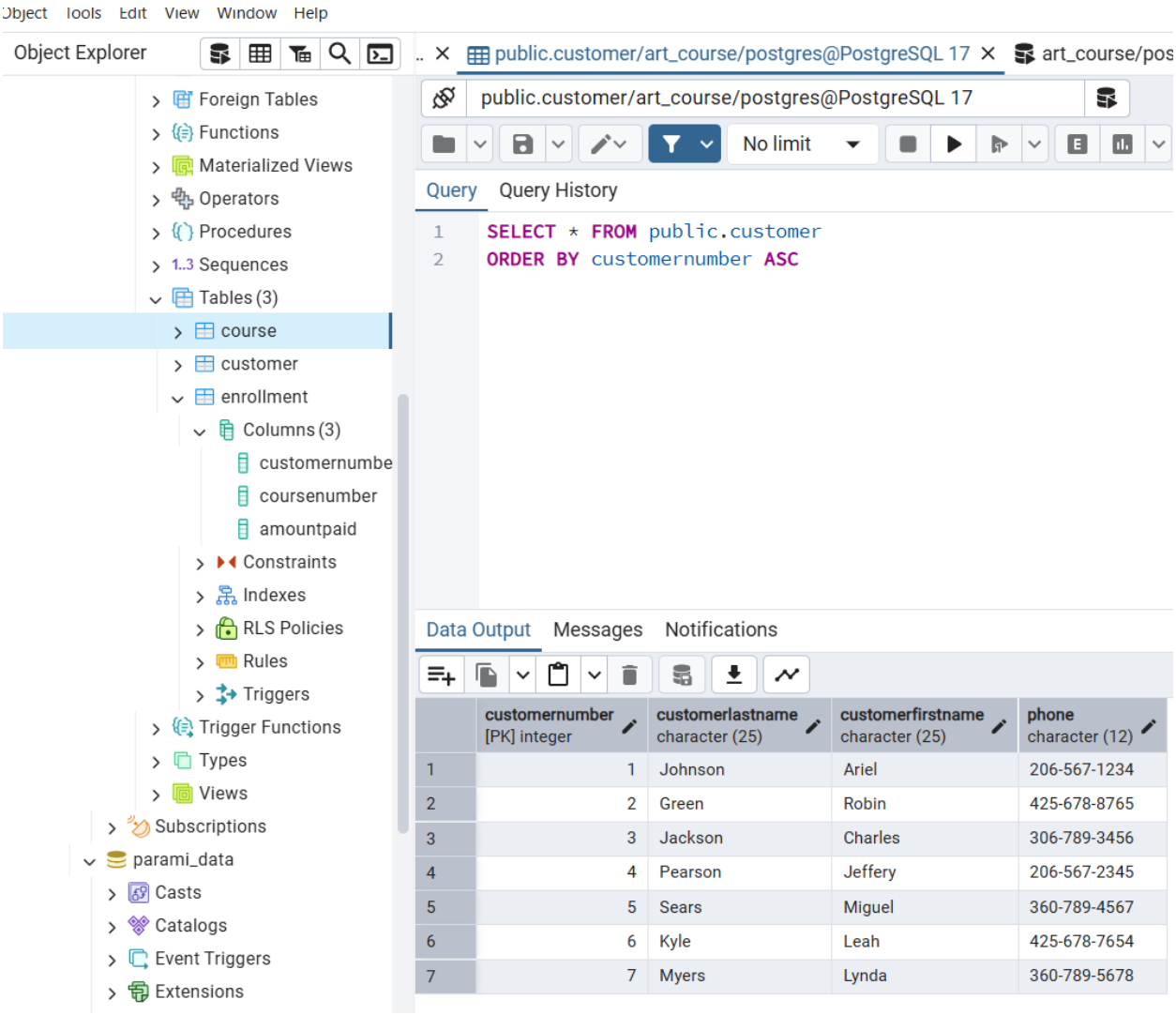


Table view:



Course Table:

Query:

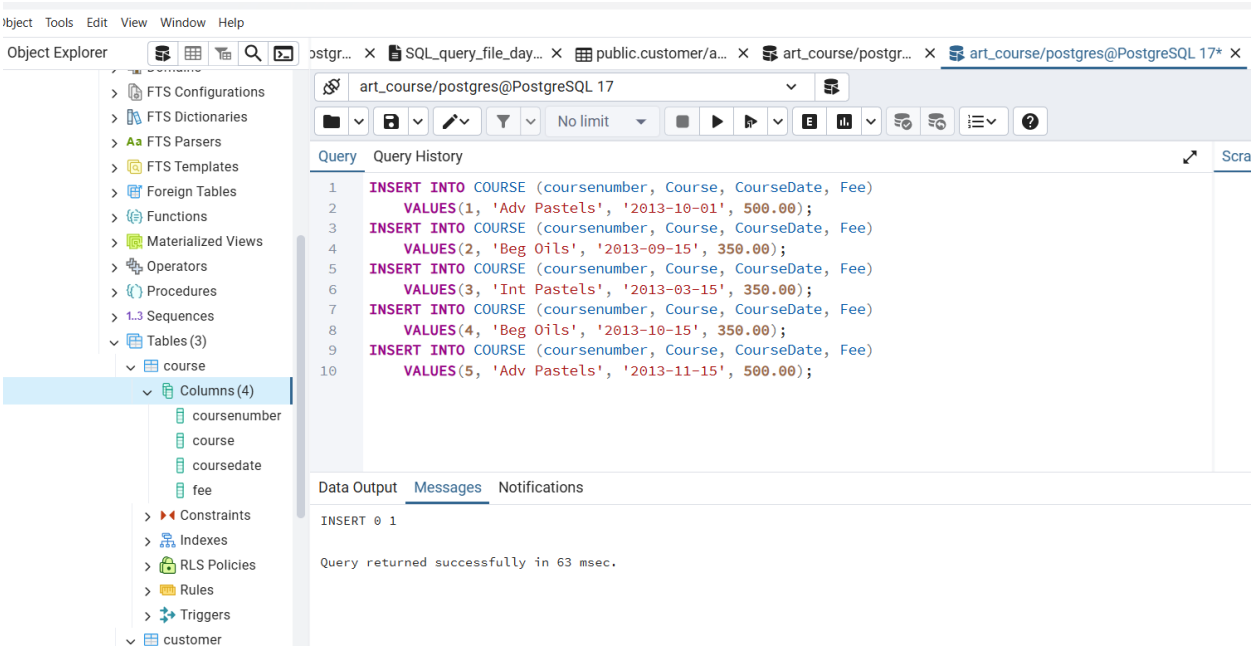
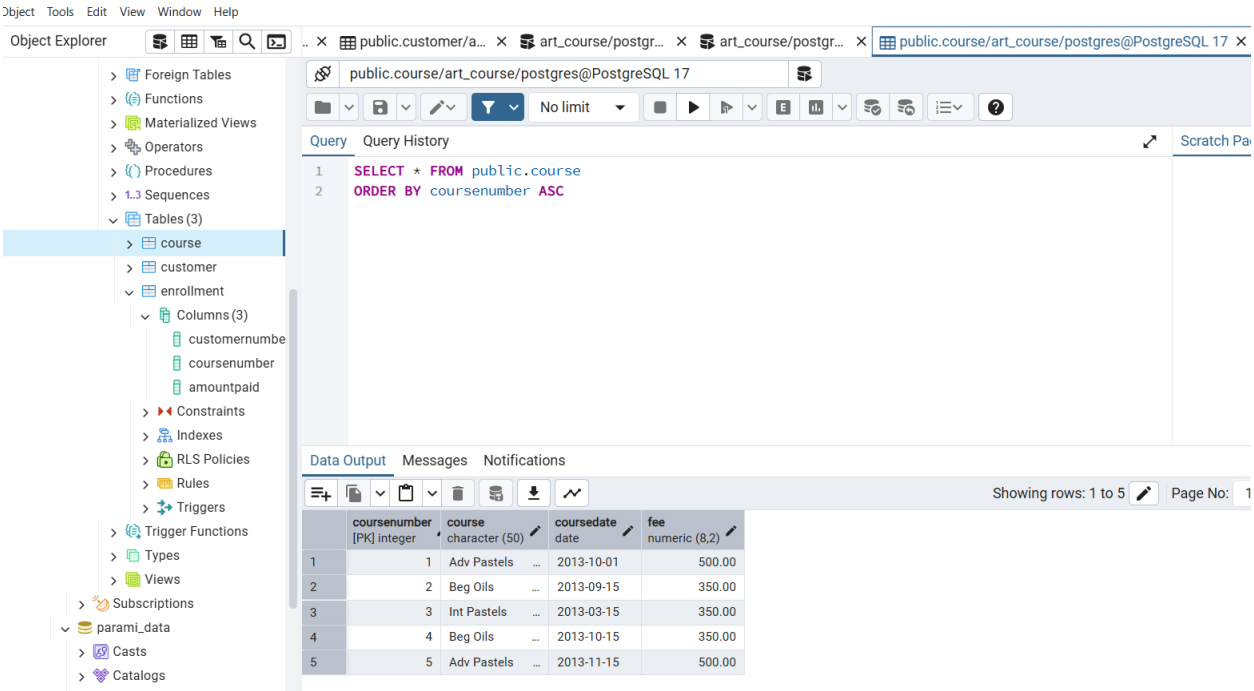

Object Explorer
art_course/postgres@PostgreSQL 17*
art_course/postgres@PostgreSQL 17
No limit
Query
Query History
INSERT INTO COURSE (coursenumber, Course, CourseDate, Fee)
VALUES(1, 'Adv Pastels', '2013-10-01', 500.00);
INSERT INTO COURSE (coursenumber, Course, CourseDate, Fee)
VALUES(2, 'Beg Oils', '2013-09-15', 350.00);
INSERT INTO COURSE (coursenumber, Course, CourseDate, Fee)
VALUES(3, 'Int Pastels', '2013-03-15', 350.00);
INSERT INTO COURSE (coursenumber, Course, CourseDate, Fee)
VALUES(4, 'Beg Oils', '2013-10-15', 350.00);
INSERT INTO COURSE (coursenumber, Course, CourseDate, Fee)
VALUES(5, 'Adv Pastels', '2013-11-15', 500.00);
Data Output
Messages
Notifications
INSERT 0 1
Query returned successfully in 63 msec.
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (3)
course
Columns (4)
coursenumber
course
coursedate
fee
Constraints
Indexes
RLS Policies
Rules
Triggers
customer

Table view:


Object Explorer
public.course/art_course/postgres@PostgreSQL 17
No limit
Query
Query History
SELECT * FROM public.course
ORDER BY coursenumber ASC
Data Output
Messages
Notifications
Showing rows: 1 to 5
Page No:
coursenumber [PK] integer
course character (50)
coursedate date
fee numeric (8,2)
1 1 Adv Pastels 2013-10-01 500.00
2 2 Beg Oils 2013-09-15 350.00
3 3 Int Pastels 2013-03-15 350.00
4 4 Beg Oils 2013-10-15 350.00
5 5 Adv Pastels 2013-11-15 500.00
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (3)
course
customer
enrollment
Columns (3)
customernumbe
coursenumber
amountpaid
Constraints
Indexes
RLS Policies
Rules
Triggers
Trigger Functions
Types
Views
Subscriptions
parami_data
Casts
Catalogs

Enrollment Table:

Query:

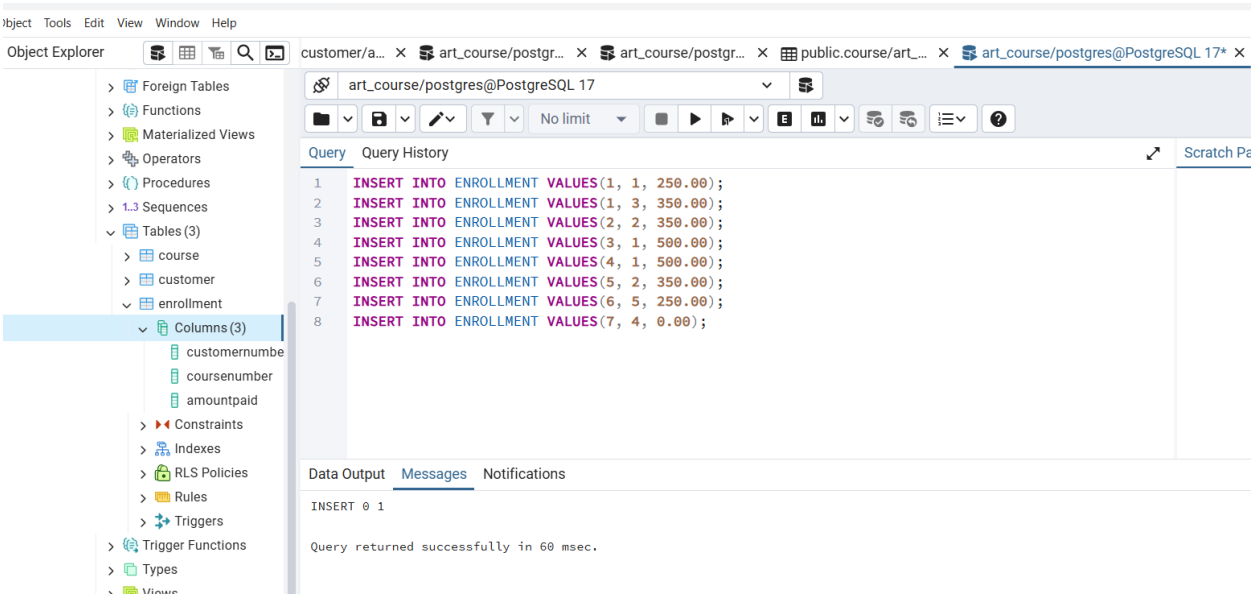
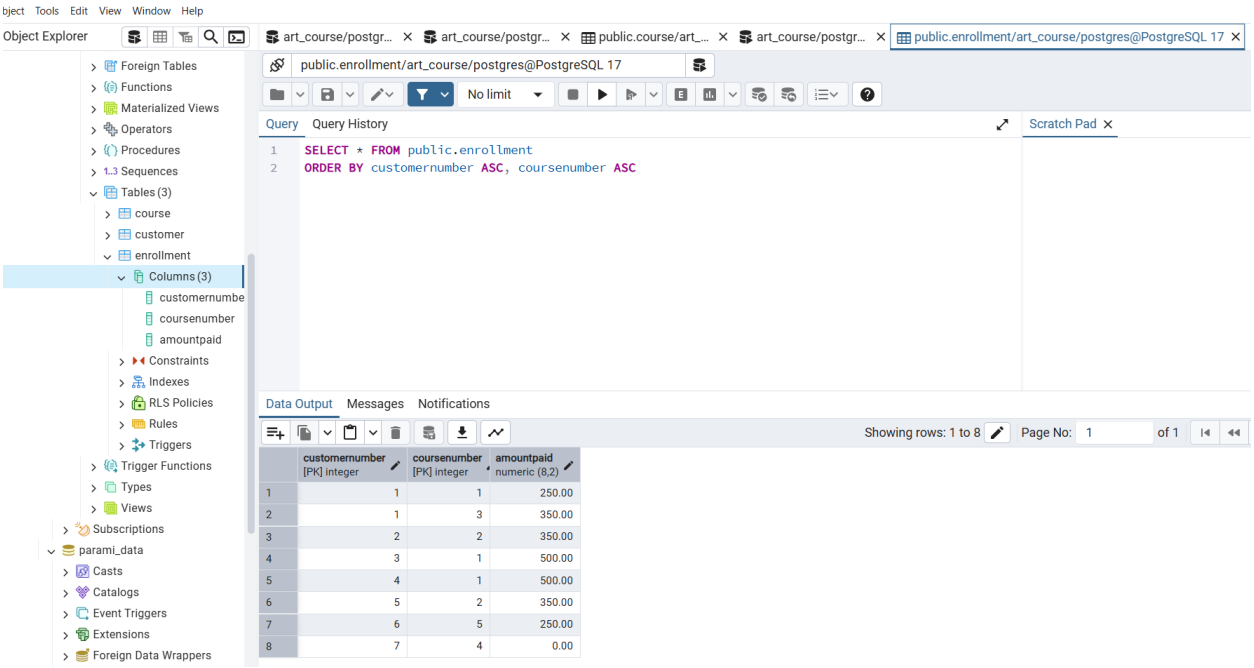


Table view:



5.

2. Using Art_Course database, answering the following questions:

2.1 How many relations are there in the Art_Course database?

2.2 List the degree and cardinality of all relations

2.3 List the primary keys and foreign keys of all relations

2.1 There are three relations/ tables in the Art_Course database. They are course table, customer table, and enrollment table.

2.2

Course table:

degree - number of columns: 4

cardinality - number of rows: 5

Customer table:

degree - number of columns: 4

cardinality - number of rows: 7

Enrollment table:

degree - number of columns: 3

cardinality - number of rows: 8

2.3

Course table:

Primary key: course_number

Foreign keys: none

Other attributes: course, fee, coursedate

Customer table:

Primary key: customer_number

Foreign keys: none

Other attributes: customerlastname, customerfirstname, phone

Enrollment table:

Primary key/Composite Key: (customer_number, course_number)

Foreign keys: course_number, customer_number

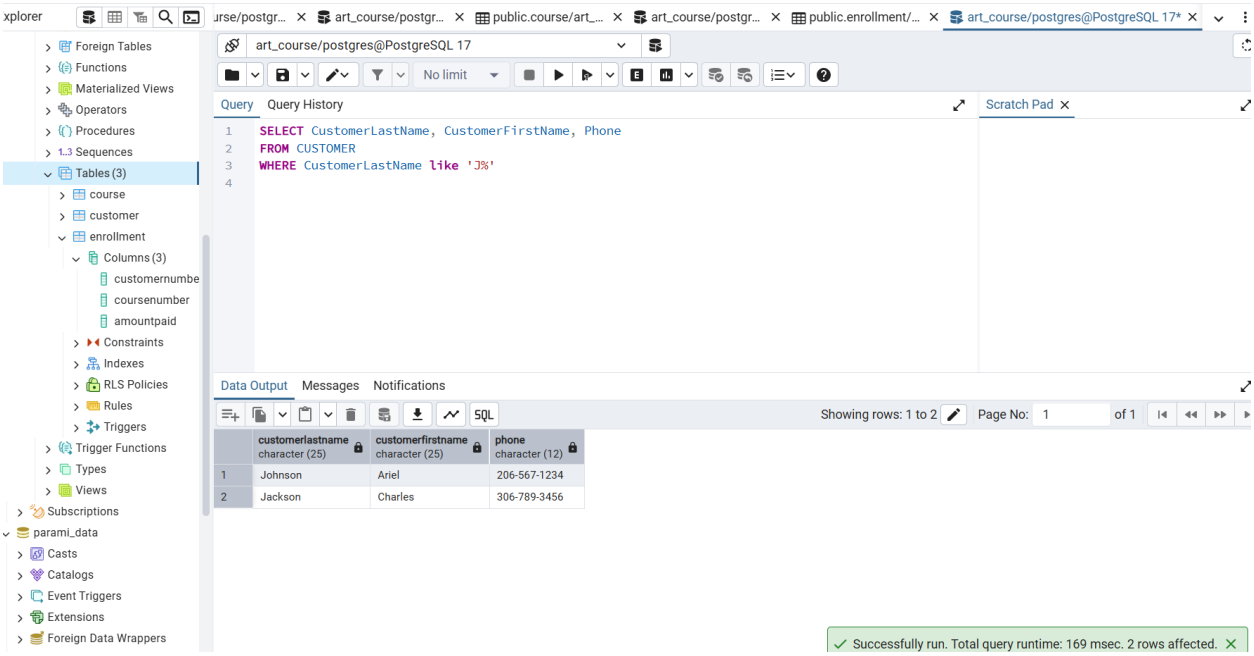
Other attributes: amount_paid

##

2.4 Using the following SQL statement, show and discuss the results

```
SELECT CustomerLastName, CustomerFirstName, Phone
FROM CUSTOMER
WHERE CustomerLastName like 'J%'
```

After running the following SQL statement:



Discussion based on the result:

According to my understanding, we select three things, CustomerLastName, CustomerFirstName, and Phone, from the Customer table. In this case, there is a condition that CustomerLastName starts with “J”. Therefore, as a result, we are getting only two rows/ two results for the customer data. The first row is about “Johnson” and the second row is about “Jackson”. Since we are selecting three columns, we are also getting their firstname and phone number as well. So, for the first row, we get “Johnson” as the CustomerLastName, “Ariel” as the CustomerFirstName, and “206567-1234” as phone. Similarly, for the second row, we get “Jackson” as the CustomerLastName, “Charles” as the CustomerFirstName and “3067893456” as phone. To conclude, I understand that this is a kind of sorting and managing the data.